

A Fine Synchronization Method for Coherent Fast Frequency Hopping

Xuanhe Yang¹, Weike Zhang², Shixun Luo³, Chang Li³, Xiaqing Miao^{3,*}, Aihua Wang¹

¹ School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China

² 96901 People's Liberation Army (PLA) Troops, Beijing 100000, China

³ School of Cyberspace Science and Technology, Beijing Institute of Technology, Beijing 100081, China

* The corresponding author, email: xqmiao@bit.edu.cn

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Abstract: Coherent fast frequency hopping (CFFH) is attracting growing attention owing to its good anti-jamming performance and the coherent combining ability. However, compared with the conventional non-coherent fast frequency hopping, CFFH requires a more precise system synchronization. In this paper, we propose a new fine synchronization algorithm for CFFH. This algorithm consists two stages, namely, open-loop stage and closed-loop stage. In the open-loop stage, a grid-based search parameter estimation method is proposed. In the closed-loop stage, we construct a fully coherent phase-locked loop (PLL) and a delay-locked loop (DLL) with decoding feedback structure to perform further fine estimation of the system clock skew and time delay, respectively. Moreover, we analyze the effect of the search parameter settings on the estimation error and derive the root mean squared error (RMSE) of estimates in the steady state of the closed-loop stage. Finally, through simulation, the RMSE performance are compared with the corresponding Cramer-Rao low bound (CRLB) and conventional code loop estimation to show the effectiveness of proposed algorithm.

Keywords: fast frequency hopping (FFH); spread spectrum communication; fine synchronization

I. INTRODUCTION

As one of the most important physical-layer secure transmission techniques, fast frequency hopping (FFH) is known for its significant low interception probability (LPI) and anti-jamming capabilities, which has been widely adopted in numerous applications [1]. Typically, the conventional FFH systems employ a non-coherent modulation and combining technique for the sake of simple implementation [2–5]. However, the signal-to-noise ratio (SNR) of each hop decreases as the frequency diversity increases, the non-coherent combination introduces a rapidly increasing non-coherent loss at the hop-level, and thus significantly deteriorates the receiver sensitivity performance. Consequently, the large diversity gain for anti-jamming reception and high receiving sensitivity for low SNR reception become a pair of trade-off, which limits the application of FFH systems. In contrast to the non-coherent FFH, coherent FFH (CFFH) scheme was proposed in the 1990s by L. Milstein [6–8]. In such scheme, the signal possesses the coherent inter-hop carrier phase, and the receiver can resort to a coherent combining of hops. Recently, CFFH is gaining renewed attention thanks to the fast development of direct digital synthesis (DDS) chips [9, 10] and new channel estimation methods [11–13]. Some new application scenarios for CFFH have been explored, such as tracking, telemetry and control (TT&C) [14], un-

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manned combat aerial vehicles (UCAV) swarms [15], industrial internet of things (IIoT) [16]. Also, some new implementations and experiments have been proposed in [17, 18].

To achieve successful frequency hopping, the transmitter and the receiver must be tightly synchronized, especially in the FFH systems [19]. In other words, synchronization techniques are crucial in FFH communications. Compared with the conventional non-coherent FFH systems, CFFH technique requires higher estimation precision of time delay and frequency deviation due to the coherent combining of multiple frequency channels [8, 6, 20]. Typically, the synchronization process for frequency hopping systems consists of two stages, i.e., coarse synchronization and fine synchronization [21]. In the coarse synchronization stage, symbol-level system synchronization is completed by the receiver, which includes the synchronization of frequency hopping patterns and coarse estimation of time delay and frequency deviation. In the fine synchronization stage, the receiver performs fine estimation of the signal time delay, frequency deviation, and carrier phase based on the output of coarse synchronization. The coarse synchronization methods of CFFH have been well investigated in many studies, for instance [21–24], whereas there are only a few studies conducted on the fine synchronization of CFFH.

Conventional algorithms for fine synchronization of signals are divided into two main categories [25]: open-loop and closed-loop algorithms, depending on the implementation structure. Open-loop algorithms have the advantage of fast estimation, but poor estimation accuracy and adaptation to parameter changes; closed-loop algorithms, on the contrary, have high accuracy and can track parameter changes, but convergence time is usually long. [26] proposed an open-loop synchronization method for hybrid CFFH and direct sequence (CFFH/DS) spread spectrum system. This approach performed parameter search in the two-dimensional plane of Doppler and code delay, and eliminated the Doppler frequency dispersion (DFD) between different frequencies by using keystone transform. In [27], an estimation method based on the randomness of multi-symbol frequency hopping patterns was proposed for the problem of integer ambiguity in CFFH/DS Doppler estimation when the number of hops was limited. It should be noted that the algo-

rithms in [26] and [27] employed differential coherent accumulation at the symbol-level and hop-level, respectively, which would inevitably cause a loss of SNR. For closed-loop algorithms, a CFFH/DS tracking system based on delay-locked loop (DLL) and phase-locked loop (PLL) was designed in [8, 6]. However, only the case of one hop in a symbol was considered, and the authors provided few insights of combining characteristics among inter-symbol hops. A non-coherent DLL-based time delay tracking structure for CFFH/DS with multi-hop in a symbol was proposed in [14], nevertheless, frequency deviation was not considered in that structure, which may not be suitable for practical implementation. In addition, all the above studies utilized the code-word correlation property of DSSS (direct sequence spread spectrum) in time synchronization, which was not applicable to non-DSSS-aided CFFH systems, and thus there was a generalization problem.

Regarding above issues, in this paper, we propose a novel fine synchronization algorithm for CFFH systems. The proposed algorithm is suitable for scenarios where the hop rate is much higher than the symbol rate, i.e., multiple hops within one symbol duration. Moreover, the proposed algorithm also allows accurate tracking of the frequency deviation generated by clock skew. The main contributions of this paper are summarized as follows:

1. First, a novel two-stage clock skew and time delay estimation method is proposed for CFFH systems. In the open-loop stage, a coherent grid search method based on the maximum likelihood criterion is proposed. Then, the effects of the search parameters on the residual estimate error are analyzed. In the closed-loop estimation stage, we develop a DLL structure to estimate/track the frequency deviation and carrier phase. Then, the received time delay is finely estimated based on the proposed DLL structure. Besides, to achieve coherent accumulation at the symbol-level, a decoding feedback scheme is employed to depolarize the symbols of the loop input during the closed-loop stage. For each stage in the algorithm, this paper presents detailed structures of practical implementations.
2. Second, we innovatively exploit the characteristics of phase differences among CFFH signal car-

riers to design the DLL structure. Different from conventional DLLs that based on the phase discriminator of time-domain correlated waveforms in a single carrier [28–30], the proposed DLL structure utilizes the phase differences of time delay among different hops to construct the early and late branches. Therefore, the phase discrimination process no longer depends on the time domain waveform on a single hop as in CFFH/DS but on the phase of all frequency channels. Consequently, the delay estimation accuracy of the system is proportional to the overall frequency hopping bandwidth.

3. Finally, the overall performance of the proposed algorithm is analyzed and extensive simulations are conducted. For the open-loop stage, the performance comparison of the correct detection probability and the root mean square error (RMSE) of estimates with different search parameters are presented. For the closed-loop stage, the RMSE of the estimates in the steady state is also derived and verified in the simulations. To verify the effectiveness of the proposed algorithm, the Cramer-Rao lower bound (CRLB) of the corresponding parameter estimation is also derived as a benchmark. The approximate expressions of the CRLB and the RMSE in the steady state are given by numerical calculations.

The rest of the paper is organized as follows. In Section II, we introduce the system model of CFFH with clock skew and time delay. Subsequently, the parameter estimation method based on grid-based open-loop searching is proposed in Section III. In Section IV, we propose the fine estimation method in closed-loop stage and the detailed structure of the loop implementation. Moreover, the RMSE of estimates is also derived in Section IV. In Section V, we derive the CRLB of parameter estimation of the CFFH system. In Section VI, simulation results of the proposed algorithm are presented. Finally, Section VII concludes this paper.

Notations: We use bold uppercase and lowercase letters to represent matrices and vectors, respectively. And $(\cdot)^T$, $(\cdot)^H$, $(\cdot)^{-1}$ denote the transpose, conjugate transpose (Hermitian), and inverse, respectively. We use $\mathbb{E}\{\cdot\}$ to denote expectation, $\Re[\cdot]$ to denote real part and $\Im[\cdot]$ to denote imaginary part. Given a matrix

K	Number of available frequency channels (number of hops per symbol).
Δf	Space between two adjacent channels in frequency hopping set.
T_h	Hop duration.
T_s	Symbol duration.
Q_{pl}	Length of pilot sequence in open-loop stage.
Q_d	Length of segment in closed-loop stage.

$\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_N] \in \mathbb{C}^{M \times N}$, we use $[\mathbf{A}]_{m,n}$ to denote the (m, n) th element of $\mathbf{A}$. We also use $[\mathbf{A}]_{m,:}$ and $[\mathbf{A}]_{:,n}$ to denote the m th row and n th column of $\mathbf{A}$ respectively. Besides, we define the vectorization operation of $\mathbf{A}$ as $\text{vec}(\mathbf{A}) = [\mathbf{a}_1^T, \mathbf{a}_2^T, \mathbf{a}_3^T, \dots, \mathbf{a}_N^T]^T$. The symbol $\odot$ stands for the hadamard product and $\lfloor x \rfloor$ means the rounding down of x . $\text{diag}(\mathbf{x})$ represents the diagonal matrix with the diagonal specified by vector $\mathbf{x}$. In this paper, $\mathbf{1}_N$ and $\mathbf{0}_N$ are used to represent all-one column vector and all-zero column vector of length N , respectively, and $\mathbf{x} = [M : N]$ denotes the row vector from integer M to integer N in steps of 1. Furthermore, $\text{sinc}(x)$ denotes the sampling function, which is defined as $\text{sinc}(x) = \begin{cases} \frac{\sin(\pi x)}{\pi x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, and $\delta(x) = \begin{cases} 0, & x \neq 0 \\ 1, & x = 0 \end{cases}$ denotes the impulse function.

For facilitate reading, some frequently parameters representation of system are listed as following:

II. SYSTEM MODEL

In this work, we consider a CFFH system [12] with K ($K \gg 1$) available frequency channels and zero intermediate frequency (ZIP). Let $\mathbf{f}_{\mathcal{M}} = [f_1, \dots, f_k, \dots, f_K]^T \in \mathbb{R}^{K \times 1}$ denotes the frequency channels, in which

$$f_k = f_1 + (K - 1)\Delta f, f_1 + \frac{K - 1}{2}\Delta f = 0, \quad (1)$$

where Δf represents the space between two adjacent frequency channels in $\mathbf{f}_{\mathcal{M}}$. To achieve the maximum frequency-domain diversity [31], we assume that each symbol traverses all K frequencies of $\mathbf{f}_{\mathcal{M}}$ and occurs once at each frequency, and under this assumption, K also denotes the number of hops per sym-

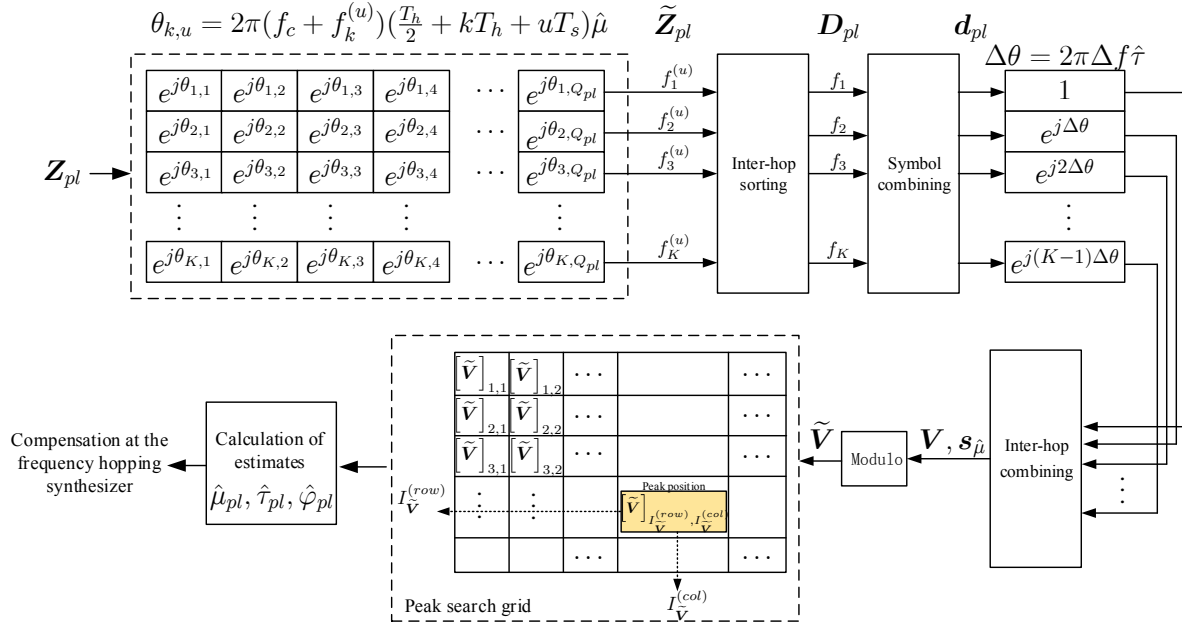


Figure 2. Block diagram of open-loop stage.

chronization [21], i.e.,

$$|\mu| < \frac{1}{2f_c T_s}, \quad |\tau| < \frac{T_h}{2M}, \quad (7)$$

where the integer $M > 1$ corresponding to the number of code slices per hop in [21].

It is worth noting that the DS is not employed in this paper. However, the coarse synchronization can be done according to the CFFH/DS systems in [21]. Without loss of generality, we assume $\Delta f = \frac{M}{T_h} = MKR_s$. Then

$$\begin{aligned} f_k^{(u)} \mu T_h &\xrightarrow{f_K \ll f_c} 0, \\ f_k^{(u)} \mu \tau &\xrightarrow{f_K \ll f_c} 0, \\ f_c \frac{\mu \tau}{2} &\xrightarrow{f_K \ll f_c, K \gg 1} 0. \end{aligned} \quad (8)$$

The received signal in (6) can be rewritten as

$$r_k^{(u)} = b^{(u)} A_{\tau, \mu, \varphi} e^{-j2\pi(f_c + f_k^{(u)})(\frac{T_h}{2} + kT_h + uT_s)\mu} e^{-j2\pi f_k^{(u)} \tau}, \quad (9)$$

where $\varphi = 2\pi f_c \tau + \varphi_0$ and $A_{\tau, \mu, \varphi} = A \left(1 - \frac{|\tau|}{T_h}\right) e^{-j\varphi} \text{sinc}(f_c \mu T_h)$ is independent of

u and k . Then, the u th symbol reception vector can be expressed as

$$\mathbf{z}^{(u)} = \mathbf{r}^{(u)} + \mathbf{n}^{(u)}, \quad (10)$$

where $\mathbf{r}^{(u)} = [r_1^{(u)}, r_2^{(u)}, \dots, r_k^{(u)}, \dots, r_K^{(u)}]^T \in \mathbb{C}^{K \times 1}$. $\mathbf{n}^{(u)}$ is the zero-mean i.i.d. additive Gaussian noise vector, which satisfies $\mathbf{n}^{(u)} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$.

III. GRID-BASED OPEN-LOOP SEARCH

In this section, we propose a parameter estimation method for μ , τ , and φ based on pilot-aid open-loop search. Without loss of generality, we assume that $b^{(u)} = 1$ for the pilot sequence and the length of the pilot sequence is Q_{pl} . Then the reception matrix is given by

$$\mathbf{Z}_{pl} = [\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \dots, \mathbf{z}^{(u)}, \dots, \mathbf{z}^{(Q_{pl})}] \in \mathbb{C}^{K \times Q_{pl}}. \quad (11)$$

The reception matrix for the signal components is expressed as

$$\mathbf{R}_{pl} = [\mathbf{r}^{(1)}, \mathbf{r}^{(2)}, \dots, \mathbf{r}^{(u)}, \dots, \mathbf{r}^{(Q_{pl})}] \in \mathbb{C}^{K \times Q_{pl}},$$

$$\left[\mathbf{r}^{(u)} \right]_k = A_{\tau, \mu, \varphi} e^{-j2\pi(f_c + f_k^{(u)})(\frac{T_h}{2} + kT_h + uT_s)\mu} e^{-j2\pi f_k^{(u)}\tau}. \quad (12)$$

Based on the maximum likelihood criterion and Gaussian distribution property of $\mathbf{n}^{(u)}$, the likelihood function estimated for μ, τ, φ can be written as

$$L(\mu, \tau, \varphi) = \left(\frac{1}{\pi\sigma_n^2} \right)^{Q_{pl}} e^{(-\|\mathbf{z}_{pl} - \mathbf{r}_{pl}\|_2^2)}, \quad (13)$$

where $\mathbf{z}_{pl} = \text{vec}(\mathbf{Z}_{pl}) \in \mathbb{C}^{KQ_{pl} \times 1}$ and $\mathbf{r}_{pl} = \text{vec}(\mathbf{R}_{pl}) \in \mathbb{C}^{KQ_{pl} \times 1}$. By simplifying (13), it is evident that maximizing $L(\mu, \tau, \varphi)$ is equivalent to searching for the μ, τ, φ to maximize the following function

$$L'(\mu, \tau, \varphi) = \Re[\mathbf{z}_{pl}^H \mathbf{r}_{pl}]. \quad (14)$$

Noting that in (12), μ and τ are relatively independent within a symbol. Moreover, φ only imposes a fixed phase rotation on signal. The estimation process can be accessed by the grid-based search shown in Figure 2. The detailed steps are as follows:

Step 1: Phase compensation for $\mathbf{Z}_{pl}$ based on an initial estimate $\hat{\mu}$. In order to perform coherent accumulation, the phase of each element in $\mathbf{Z}_{pl}$ should be aligned. According to (12), the received phase of each hop is affected by μ and τ , and here the compensation is performed for μ first.

Denote the $\mathbf{Z}_{pl}$ phase compensation result as $\tilde{\mathbf{Z}}_{pl} \in \mathbb{C}^{K \times Q_{pl}}$. The $\tilde{\mathbf{Z}}_{pl}$ can be expressed as

$$\tilde{\mathbf{Z}}_{pl} = e^{2\pi(f_c + f_k^{(u)})\frac{T_h}{2}\hat{\mu}} \text{diag} \left(e^{(j2\pi(f_c + f_k^{(u)})[1:K]^T \hat{\mu})} \right) \cdot \mathbf{Z}_{pl} \text{diag} \left(e^{(j2\pi(f_c + f_k^{(u)})[1:Q] \hat{\mu})} \right). \quad (15)$$

The initial estimate can be set as $\hat{\mu} = \hat{\mu}_1 = -\frac{1}{2f_c T_s}$ to cover the range of residual errors in the coarse synchronization. As shown in Figure 2, the phase compensation of (15) corresponding to the u th symbol k th hop is

$$\theta_{k,u} = 2\pi(f_c + f_k^{(u)}) \left(\frac{T_h}{2} + kT_h + uT_s \right) \hat{\mu}. \quad (16)$$

Step 2: Since the actual τ is unknown, the accumulation and combining process should be firstly performed in symbol-by-symbol manner and then hop-by-hop. To facilitate the processing, we first sort the intra-symbol hops of $\tilde{\mathbf{Z}}_{pl}$ in order of frequency from lowest to highest to obtain $\mathbf{D}_{pl} \in \mathbb{C}^{K \times Q_{pl}}$, i.e.,

$$[\mathbf{D}_{pl}]_{k,u} = [\tilde{\mathbf{Z}}_{pl}]_{\Psi_u^{(-1)}(k),u}. \quad (17)$$

By accumulating the rows of $\mathbf{D}_{pl}$, we can get the symbol-level accumulation result $\mathbf{d}_{pl} = \mathbf{D}_{pl} \mathbf{1}_{Q_{pl}} \in \mathbb{C}^{K \times 1}$.

Step 3: The main object of this step is to compensate the inter-hop phase difference brought by the τ . Since Δf is fixed, after frequency ordering, τ will correspond to the linear changes of the intra-symbol inter-hop phases. Specifically, the phase difference of adjacent elements of $\mathbf{d}_{pl}$ is $\Delta\theta = 2\pi\Delta f\tau$. With this property, the τ compensation process can be performed using the IFFT. The result of hop-level combining is expressed as

$$\mathbf{s}_{\hat{\mu}_1} = \frac{1}{U} \text{fftshift}(\mathbf{F}_U^H [\mathbf{d}_{pl}^T, \mathbf{0}_{U-K}^T]^T) \in \mathbb{C}^{U \times 1}, \quad (18)$$

where $\mathbf{F}_U$ is the $U \times U$ Fourier matrix and function $\text{fftshift}(\mathbf{x})$ rearranges a Fourier transform $\mathbf{x}$ by shifting the zero-frequency component to the center of the vector. $\mathbf{0}_{U-K}$ denotes the complementary zero to the estimated vector $\mathbf{d}_{pl}$. The compensation range corresponding to $\mathbf{s}_{\hat{\mu}_1}$ is $\left[-\frac{1}{2\Delta f}, \frac{1}{2\Delta f} - \frac{1}{U\Delta f} \right]$, which covers the error range in (7), and the compensation precision for τ is $\Delta\tau_{pl} = \frac{1}{U\Delta f} = \frac{T_h}{UM}$.

Step 4: Repeat the adjustment of the search parameters in Step 1 to form the search dimension for μ . Update $\hat{\mu}$ with fixed step $\Delta\mu_{pl} = \frac{1}{NQ_{pl}f_c T_s}$, where N is a predefined parameter that controls the search precision. The i th compensation of μ can be expressed as

$$\hat{\mu}_i = \hat{\mu}_1 + (i-1)\Delta\mu_{pl}$$

$$= \frac{2i - 2 - NQ_{pl}}{2NQ_{pl}f_cT_s}, i = 1, \dots, NQ_{pl} + 1. \quad (19)$$

Repeat Step 1–4 and stack $\mathbf{s}_{\hat{\mu}_i}$ into a record matrix $\mathbf{V} = [\mathbf{s}_{\hat{\mu}_1}, \mathbf{s}_{\hat{\mu}_2}, \dots, \mathbf{s}_{\hat{\mu}_{NQ_{pl}+1}}] \in \mathbb{C}^{U \times (NQ_{pl}+1)}$. Then, each element in $\mathbf{V}$ represents a coherent accumulation and combining result of $\mathbf{Z}_{pl}$ with the corresponding $\hat{\mu}$, $\hat{\tau}$ compensation.

Step 5: Modulo each element of $\mathbf{V}$ to obtain the search grid $\tilde{\mathbf{V}} \in \mathbb{C}^{U \times (NQ_{pl}+1)}$, and find the maximum coordinate $(I_{\tilde{\mathbf{V}}}^{(row)}, I_{\tilde{\mathbf{V}}}^{(col)})$ in $\tilde{\mathbf{V}}$ by traversing. Then, the estimates of μ and τ correspond to $I_{\tilde{\mathbf{V}}}^{(col)}$ and $I_{\tilde{\mathbf{V}}}^{(row)}$, respectively, and the estimate of φ corresponds to the phase of the $[\mathbf{V}]_{I_{\tilde{\mathbf{V}}}^{(row)}, I_{\tilde{\mathbf{V}}}^{(col)}} e^{j2\pi f_1 \hat{\tau}_{pl}}$, as shown in (20).

$$\begin{aligned} \hat{\mu}_{pl} &= \frac{2I_{\tilde{\mathbf{V}}}^{(col)} - 2 - NQ_{pl}}{2NQ_{pl}f_cT_s}, \\ \hat{\tau}_{pl} &= \frac{I_{\tilde{\mathbf{V}}}^{(row)} - U/2 - 1}{U\Delta f}, \\ \hat{\varphi}_{pl} &= \arctan 2 \left([\mathbf{V}]_{I_{\tilde{\mathbf{V}}}^{(row)}, I_{\tilde{\mathbf{V}}}^{(col)}} e^{j2\pi f_1 \hat{\tau}_{pl}} \right), \end{aligned} \quad (20)$$

where $\arctan 2$ is the four-quadrant arctangent function. Finally, the results in (20) are fed back to the frequency hopping synthesizer to compensate the time delay, clock skew, and the initial phase.

Next, we investigate the peak shape of the search grid $\tilde{\mathbf{V}}$. First, we focus on the dimension of τ . Assuming the error of μ equal to 0, $\mathbf{d}_{pl}$ can be expressed as

$$\mathbf{d}_{pl} = Q_{pl} A_{\tau, \mu, \varphi} e^{-j2\pi f_{\mathcal{M}} \tau}. \quad (21)$$

Consider the case that the $\hat{\tau}_{pl}$ is close to the actual τ , the value of the grid point after IFFT can be approximated as

$$\begin{aligned} &\sum_{k=1}^K Q_{pl} A_{\tau, \mu, \varphi} e^{-j2\pi(f_k \tau - \frac{x(k-1)}{U})} \\ &= A_{\tau, \mu, \varphi} Q_{pl} e^{-j2\pi(f_1 \tau + (\Delta f \tau - \frac{x}{U}) \frac{K-1}{2})} \end{aligned}$$

$$\begin{aligned} &\cdot \frac{\sin(\pi(\Delta f \tau - \frac{x}{U})K)}{\sin(\pi(\Delta f \tau - \frac{x}{U}))} \\ &\approx A_{\tau, \mu, \varphi} Q_{pl} K e^{j2\pi \frac{x}{U} \frac{K-1}{2}} \text{sinc} \left((\Delta f \tau - \frac{x}{U})K \right) \\ &= A_{\tau, \mu, \varphi} Q_{pl} K e^{-j2\pi f_1 \hat{\tau}_{pl}} \text{sinc} \left((\Delta f \tau - \frac{x}{U})K \right), \end{aligned} \quad (22)$$

where $x = I_{\tilde{\mathbf{V}}}^{(row)} - U/2 - 1$ is the τ -dimension coordinate corresponding to $\hat{\tau}_{pl}$ in $\tilde{\mathbf{V}}$.

For the analysis in dimension μ , similarly, the τ -dimensional estimation error is assumed to zero. Denoting the coordinate of $\hat{\mu}_{pl}$ in μ -dimension as $y = I_{\tilde{\mathbf{V}}}^{(col)} - NQ_{pl}/2 - 1$, the phase error of u th symbol k th hop is

$$\begin{aligned} \theta_{u,k}^{(error)}(y) &= 2\pi(f_c + f_k^{(u)})uT_s \left(\mu - \frac{y}{2NQ_{pl}f_cT_s} \right) \\ &\quad + 2\pi(f_c + f_k^{(u)}) \left(\frac{1}{2} + k \right) T_h \left(\mu - \frac{y}{2NQ_{pl}f_cT_s} \right). \end{aligned} \quad (23)$$

When N and Q_{pl} are relatively large, y satisfies $f_cT_s\mu - \frac{y}{2NQ_{pl}f_cT_s} \rightarrow 0$ for points that are close to actual μ in μ -estimation. In this case, the effect of the second term (T_h -related term) in equation (23) can be ignored and $\theta_{u,k}^{(error)}(y)$ mainly depends on the result of the first term. After accumulation and combining, the corresponding values in $\mathbf{V}$ can be approximated as

$$\begin{aligned} &\sum_{k=1}^K \sum_{u=1}^{Q_{pl}} A_{\tau, \mu, \varphi} e^{-j2\pi(f_c + f_k)uT_s(\mu - \frac{y}{2NQ_{pl}f_cT_s})} \\ &\approx A_{\tau, \mu, \varphi} Q_{pl} \text{sinc} \left(f_cT_s(\mu - \frac{y}{2NQ_{pl}f_cT_s})Q_{pl} \right) \\ &\quad \cdot \sum_{k=1}^K e^{-j\pi(Q_{pl}-1)(f_c + f_k)T_s(\mu - \frac{y}{2NQ_{pl}f_cT_s})} \\ &\approx A_{\tau, \mu, \varphi} Q_{pl} K \text{sinc} \left(f_cT_s(\mu - \frac{y}{2NQ_{pl}f_cT_s})Q_{pl} \right) \\ &\quad \cdot \text{sinc} \left(\frac{(Q_{pl}-1)K}{2}(\mu - \frac{y}{2NQ_{pl}f_cT_s})T_s\Delta f \right) \\ &\quad \cdot e^{-j2\pi \frac{Q_{pl}-1}{2}(\mu - \frac{y}{2NQ_{pl}f_cT_s})T_s(f_c + f_1 + \frac{K-1}{2}\Delta f)} \\ &\approx A_{\tau, \mu, \varphi} Q_{pl} K \text{sinc} \left(f_cT_s(\mu - \frac{y}{2NQ_{pl}f_cT_s})Q_{pl} \right) \\ &\quad \cdot e^{-j2\pi \frac{Q_{pl}-1}{2}(\mu - \frac{y}{2NQ_{pl}f_cT_s})f_cT_s}. \end{aligned} \quad (24)$$

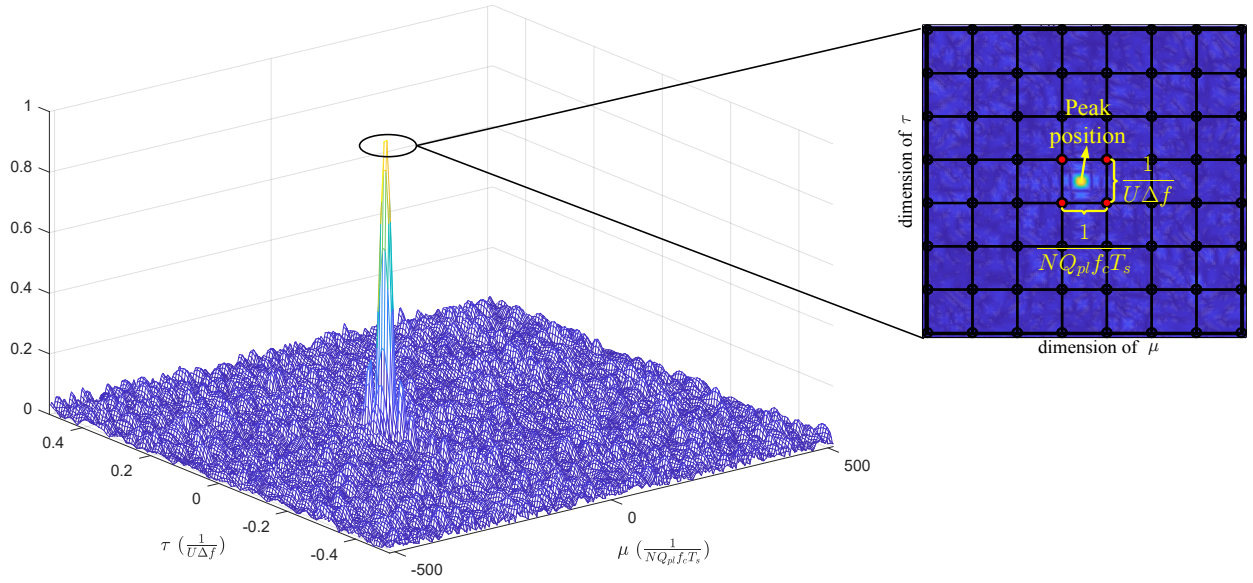


Figure 3. Peak normalized open-loop search grid, $K = 32$, $N = 8$, $U = 8$, $Q_{pl} = 128$, $E_s/N_0 = 10$ dB.

As can be seen from (23) and (24), at the peak of $\tilde{\mathbf{V}}$, the roll-off of both τ and μ dimensions is sinc shape and the main lobe widths $\frac{2U}{K}$ and $2N$, respectively. Figure 3 shows the results for $\tilde{\mathbf{V}}$ with $K = 32$, $N = 8$, $U = 8K$, $Q_{pl} = 128$, $E_s/N_0 = 10$ dB.

Define correct detection as the peak on $\tilde{\mathbf{V}}$ lies in the same grid as the true values of μ and τ . Let $\Delta\hat{\mu}_{pl}$, $\Delta\hat{\tau}_{pl}$ and $\Delta\hat{\varphi}_{pl}$ denote the estimation errors of grid-based search. Then, under the correct detection conditions, the ranges of $\Delta\hat{\mu}_{pl}$ and $\Delta\hat{\tau}_{pl}$ are

$$\begin{aligned} |\Delta\hat{\mu}_{pl}| &\leq \frac{1}{NQ_{pl}f_cT_s}, \\ |\Delta\hat{\tau}_{pl}| &\leq \frac{1}{U\Delta f}. \end{aligned} \quad (25)$$

For φ , since the peak is the result of the coherent accumulation of Q_{pl} symbols, by substituting $\Delta\hat{\mu}_{pl} = \frac{1}{NQ_{pl}f_cT_s}$ into (24), the range of $\Delta\hat{\varphi}_{pl}$ at correct detection can be written as

$$\begin{aligned} |\Delta\hat{\varphi}_{pl}| &\leq 2\pi \frac{Q_{pl} - 1}{2} \frac{1}{NQ_{pl}f_cT_s} f_cT_s \\ &= \frac{Q_{pl} - 1}{NQ_{pl}} \pi \approx \frac{\pi}{N}. \end{aligned} \quad (26)$$

From (25) and (26), it is clear that in order to minimize the estimation errors of τ , μ and φ , N and U must be selected as large as possible.

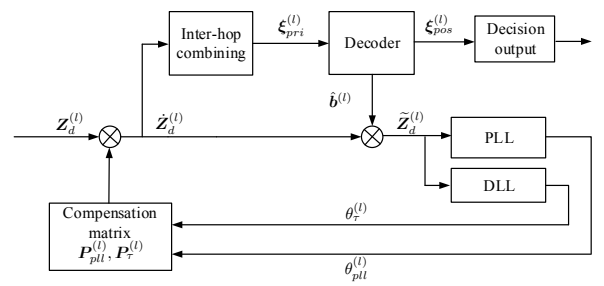


Figure 4. Block diagram of closed-loop fine parameter estimation.

IV. CLOSED-LOOP FINE PARAMETER ESTIMATION

In this section, we investigate the closed-loop fine parameter estimation of the residual time delay, clock skew, and continuous tracking of the carrier phase.

Different from Section III, the proposed fine parameter estimation in this section is based on the data segment in stead of pilot sequence. As shown in Figure 4, the proposed algorithm consists of three parts, namely, decoder, PLL and DLL. First, the decoder extracts the log-likelihood ratio (LLR) of the input data to conduct de-polarization through decoding feedback. Then, the PLL performs tracking of the signal carrier phase and the fine estimation of the clock skew. Meanwhile, the DLL performs fine estimation of the time delay. The algorithm works in an iterative form, i.e., the estimation

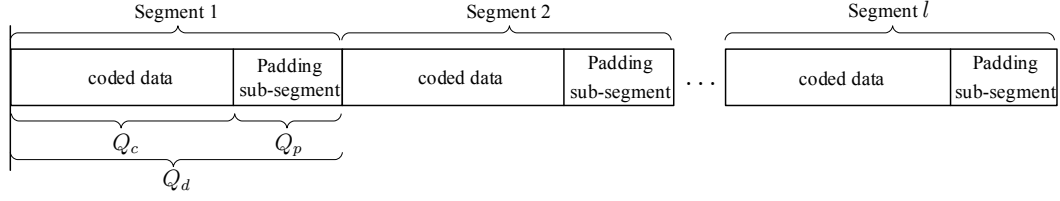


Figure 5. Data segment frame.

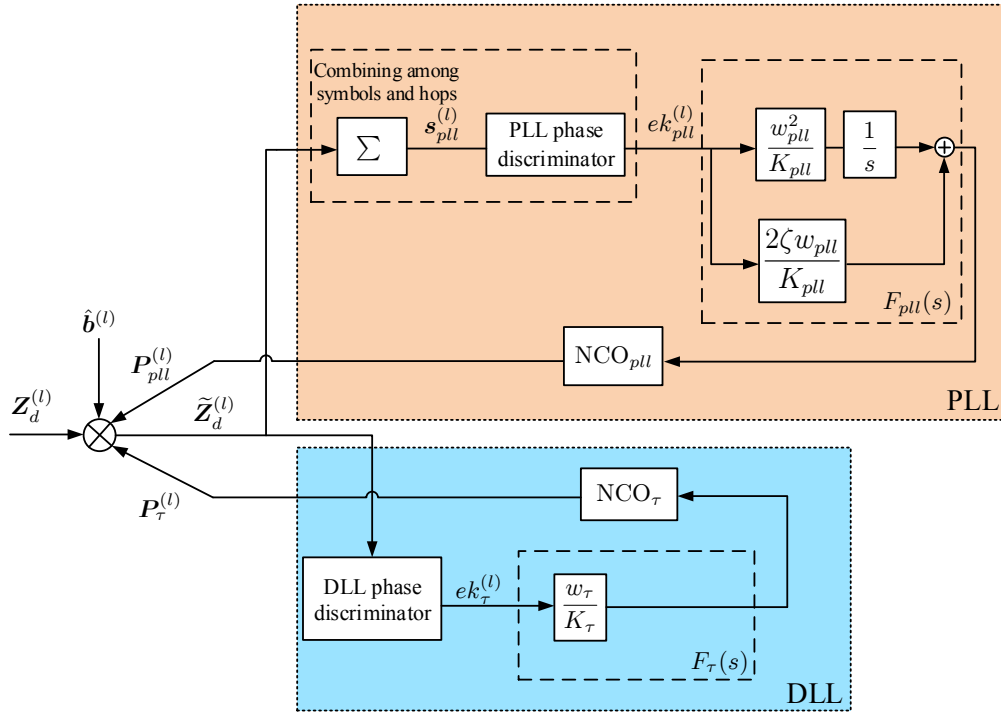


Figure 6. Block diagram of PLL and DLL feedback.

result of each segment is fed back to the frequency hopping synthesizer to compensate next segment. In addition, a all-one padding sub-segment is attached to the end of each segment to deal with the hardware latency of decoding and feedback processing at the receiver. Let $Q_d = Q_c + Q_p$ be the length of each segment, where Q_c denotes the coded data length and Q_p denotes the length of the padded sub-segment. The data segment frame is illustrated in Figure 5.

Figure 6 shows the structure of the PLL and the DLL. In the proposed PLL, second-order structure is adopted to perform fine estimation of clock skew and the tracking of small jitter. Different from PLL, the DLL in this paper uses a first-order structure to per-

form time delay estimation since there is no relative motion between the transmitter and receiver.

Let the closed-loop fine parameter estimation algorithm input start from the first symbol, then the l th iteration input matrix can be expressed as

$$\mathbf{Z}_d^{(l)} = \begin{bmatrix} z^{((l-1)Q_d+1)}, z^{((l-1)Q_d+2)}, \\ \dots, z^{((l-1)Q_d+Q_c)}, \dots, z^{(lQ_d)} \end{bmatrix} \in \mathbb{C}^{K \times Q_d}. \quad (27)$$

As shown in Figure 6, the output of the PLL and the DLL will be fed back to $\mathbf{Z}_d^{(l)}$. Denote $\dot{\mathbf{Z}}_d^{(l)} \in \mathbb{C}^{K \times Q_d}$

and $\tilde{\mathbf{Z}}_d^{(l)} \in \mathbb{C}^{K \times Q_d}$ as the compensation output and de-polarization output. Then $\dot{\mathbf{Z}}_d^{(l)}$ and $\tilde{\mathbf{Z}}_d^{(l)}$ can be expressed as

$$\begin{aligned}\dot{\mathbf{Z}}_d^{(l)} &= \mathbf{Z}_d^{(l)} \odot e^{-j2\pi \mathbf{P}_{pll}^{(l)}} \odot e^{-j2\pi \mathbf{P}_\tau^{(l)}}, \\ \tilde{\mathbf{Z}}_d^{(l)} &= \dot{\mathbf{Z}}_d^{(l)} \text{diag}(\hat{\mathbf{b}}^{(l)}),\end{aligned}\quad (28)$$

where $\mathbf{P}_{pll}^{(l)} \in \mathbb{C}^{K \times Q_d}$, $\mathbf{P}_\tau^{(l)} \in \mathbb{C}^{K \times Q_d}$ are PLL and DLL compensation matrices for the $l-1$ th iteration, and the initial values are $\mathbf{P}_{pll}^{(0)} = \mathbf{P}_\tau^{(0)} = \mathbf{O}_{K \times Q_d}$. $\hat{\mathbf{b}}^{(l)} \in \mathbb{R}^{Q_d \times 1}$ is the estimate vector of the input symbols, expressed as

$$\hat{\mathbf{b}}^{(l)} = \left[\left(\hat{\mathbf{b}}_c^{(l)} \right)^T, \mathbf{1}_{Q_p}^T \right]^T, \quad (29)$$

where $\hat{\mathbf{b}}_c^{(l)} \in \mathbb{R}^{Q_c \times 1}$ is the coded symbol estimate vector and $\mathbf{1}_{Q_p}$ is a $Q_p \times 1$ vector of padded sub-segment. To facilitate the description, in the following we assume $\mathbf{b}^{(l)} = \hat{\mathbf{b}}^{(l)} = \mathbf{1}_{Q_d}$ in Section 4.2–4.3. The detailed discussion of generating $\hat{\mathbf{b}}^{(l)}$ is presented in the Section 4.4.

4.1 Introduction of PLL Structure

Denote μ_d , τ_d , and φ_d as the initial errors of μ , τ , and φ , respectively. Recalling (25), The phase difference induced by μ_d among intra-symbol and inter-symbol hops within a segment can be written as

$$\begin{aligned}\Delta\theta(\text{intra-symbol}) &\leq 2\pi \left[(f_K + f_c) \left(K - \frac{1}{2} \right) T_h - (f_1 + f_c) \frac{1}{2} T_h \right] \mu_d \\ &\leq \frac{2\pi(f_c + \frac{K}{2}\Delta f)(K-1)T_h}{2NQ_{pl}f_cT_s} \approx \frac{(K-1)\pi}{KNQ_{pl}}, \\ \Delta\theta(\text{inter-symbol}) &\leq 2\pi f_c Q_d T_s \mu_d \leq \frac{2\pi f_c Q_d T_s}{2NQ_{pl}f_cT_s} = \frac{\pi Q_d}{NQ_{pl}}.\end{aligned}\quad (30)$$

It can be seen that Q_d should be constrained far less than NQ_{pl} so that the approximate coherent combining and accumulation can be performed among hops and symbols. In such case, the result of coherent accumulation in $Q_d T_s$ can be expressed as $s_{pll}^{(l)} =$

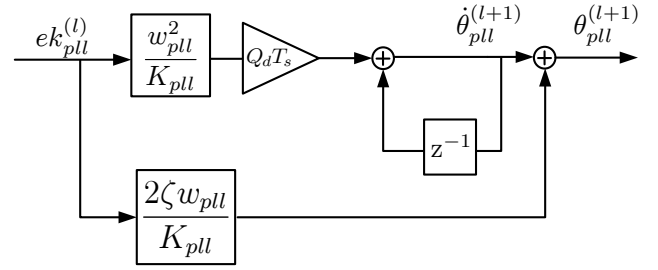


Figure 7. PLL loop filter digital implementation.

$\sum_{u=1}^{Q_d} \sum_{k=1}^K [\tilde{\mathbf{Z}}_d^{(l)}]_{k,u}$. The phase discrimination formula of the PLL is

$$ek_{pll}^{(l)} = \arctan 2 \left(\Im \left[s_{pll}^{(l)} \right], \Re \left[s_{pll}^{(l)} \right] \right), \quad (31)$$

where $ek_{pll}^{(l)}$ is the phase discrimination error output of the l th iteration.

Corresponding to Figure 6, the loop filter for the PLL can be expressed as

$$F_{pll}(s) = \frac{2\zeta w_{pll}s + w_{pll}^2}{K_{pll}s}, \quad (32)$$

where ζ is the loop damping factor and w_{pll} is the characteristic frequency. Note that $0.4 < \zeta < 0.8$ is adopted to ensure stable and fast convergence for the second-order system. K_{pll} is the total gain of PLL. In this paper, both the phase discriminator and the NCO_{pll} output the normalized result, i.e. $K_{pll} = 1$.

By the boxcar digital integrator method [32], the analog filter (32) can be mapped to the digital implementation in (33), as shown in Figure 7.

$$F_{pll}(z) = \frac{(w_{pll}^2 Q_d T_s + 2\zeta w_{pll}) - 2\zeta w_{pll} z^{-1}}{K_{pll}(1 - z^{-1})}. \quad (33)$$

The recursive expression for this loop filter can be obtained using the Jaffe-Rechtin filter algorithm [33], which is expressed as

$$\begin{aligned}\dot{\theta}_{pll}^{(l+1)} &= \dot{\theta}_{pll}^{(l)} + \frac{w_{pll}^2 \cdot Q_d T_s}{K_{pll}} \cdot ek_{pll}^{(l)}, \\ \theta_{pll}^{(l+1)} &= \dot{\theta}_{pll}^{(l+1)} + \frac{2\zeta \cdot w_{pll}}{K_{pll}} \cdot ek_{pll}^{(l)},\end{aligned}\quad (34)$$

where $\dot{\theta}_{pll}^{(1)} = 0$.

The function of NCO_{pll} is to calculate the phase compensation matrix with $\dot{\theta}_{pll}^{(l)}$. Since the hopping patterns vary with the symbols, the compensation phase for each symbol and hop in the next iteration should be pre-calculated. Let $\mathbf{P}_0^{(l+1)} \in \mathbb{C}^{K \times (Q_d+1)}$ be the compensation basis matrix for each symbol in the $l+1$ th iteration. After normalizing to T_s , $\mathbf{P}_0^{(l+1)}$ is expressed as

$$\mathbf{P}_0^{(l+1)} = \theta_{pll}^{(l+1)} \left(1 + \frac{f_M}{f_c} \right) [0 : Q_d] + [\mathbf{P}_0^{(l)}]_{:,Q_d+1} \cdot \mathbf{1}_{Q_d+1}^T, \quad (35)$$

where $\mathbf{P}_0^{(1)} = \mathbf{O}_{K,Q_d+1}$. The first Q_d columns of $\mathbf{P}_0^{(l)}$, namely $[\mathbf{P}_0^{(l)}]_{:,1:Q_d-1}$, generate the increase in phase of hops of Q_d symbols in l th iteration. The last column of $\mathbf{P}_0^{(l)}$, namely $[\mathbf{P}_0^{(l)}]_{:,Q_d}$ records the accumulation of each hop phase during $l-1$ iterations. Taking $\mathbf{P}_0^{(l)}$ as the basis, the equation of the NCO_{pll} can be expressed as

$$[\mathbf{P}_{pll}^{(l)}]_{k,u} = \begin{cases} \theta_{pll}^{(l)} \left(1 + \frac{f_k^{((l-1)Q_d+u)}}{f_c} \right) (k - \frac{1}{2}) T_h \\ + [\mathbf{P}_0^{(l)}]_{\Psi_{(l-1)Q_d+u}^{(-1)}(k),u}, & l > 1. \\ 0, & l = 1. \end{cases} \quad (36)$$

4.2 Introduction of DLL Structure

The basic principle of DLL is to construct early-late discriminator to detect time delay and filter the estimation results in the loop structure. In order to obtain an accurate estimate, the output of the discriminator should respond linearly to the actual time delay within the estimated range. The classical approach is to utilize the code loop and calculate delay by the roll-off character at auto-correlation peak of the code-word [8, 28, 29]. Different from this classical approach, the algorithm described in this section exploit the delay-phase characteristics of CFFH, and construct the early-late discriminator though the phase differences among K frequency channels.

For the convenience of description, the principle of DLL phase discrimination is first illustrated by the

combination of single symbol. For the u th symbol, the early and late branches are constructed by compensating the phase of each hop according to $\frac{f^{(u)}-f_1}{2K\Delta f}$ and $\frac{-(f^{(u)}-f_1)}{2K\Delta f}$, respectively. Denoting the time delay error as $\tau' = \frac{\lambda}{K\Delta f}$, the normalized compensation results of the two branches can be expressed as

$$\begin{aligned} ek_E &= \frac{1}{K} \sum_{k=1}^K e^{-j2\pi f_k \frac{\lambda}{K\Delta f}} e^{-j2\pi \frac{k-1}{2K}} \\ &= \frac{1}{K} \frac{\cos(\pi\lambda)}{\sin(\frac{\pi}{K}(\lambda + \frac{1}{2}))} e^{-j\pi \frac{K-1}{2K}}, \\ ek_L &= \frac{1}{K} \sum_{k=1}^K e^{-j2\pi f_k \frac{\lambda}{K\Delta f}} e^{j2\pi \frac{k-1}{2K}} \\ &= \frac{1}{K} \frac{-\cos(\pi\lambda)}{\sin(\frac{\pi}{K}(\lambda - \frac{1}{2}))} e^{j\pi \frac{K-1}{2K}}. \end{aligned} \quad (37)$$

Recalling (25), the residual error of the time delay $|\tau'| \leq \frac{1}{U\Delta f}$, then, $|\lambda| \leq \frac{K}{U} \leq 1$. The phase discriminator for a single symbol can be expressed as

$$\begin{aligned} ek_0 &= \frac{1}{2} \frac{\Re[ek_E \cdot e^{j\pi \frac{K-1}{2K}}] - \Re[ek_L \cdot e^{-j\pi \frac{K-1}{2K}}]}{\Re[ek_E \cdot e^{j\pi \frac{K-1}{2K}}] + \Re[ek_L \cdot e^{-j\pi \frac{K-1}{2K}}]} \\ &= \frac{1 \sin(\frac{\pi}{K}(\lambda - \frac{1}{2})) + \sin(\frac{\pi}{K}(\lambda + \frac{1}{2}))}{2 \sin(\frac{\pi}{K}(\lambda - \frac{1}{2})) - \sin(\frac{\pi}{K}(\lambda + \frac{1}{2}))} \\ &= -\frac{1 \tan(\frac{\pi}{K}\lambda)}{2 \tan(\frac{\pi}{2K})} \xrightarrow{K \text{ is large}} -\lambda. \end{aligned} \quad (38)$$

It can be seen that by the process of (37) and (38), we obtain a discrimination result linearly related to the time delay.

For the l th iteration, Q_d symbols with different hopping patterns are combined, and the phase discrimination process of the DLL is shown in Figure 8. The early and late branches are designed as

$$\begin{aligned} \rho_E^{(l)} &= \Re \left[\sum_{u=1}^{Q_d} \sum_{k=1}^K e^{j\pi \frac{K-1}{2K}} e^{-j\pi \frac{\Psi_{(l-1)Q_d+u}^{(-1)}(k)-1}{K}} \frac{[\tilde{\mathbf{Z}}_d^{(l)}]_{k,u}}{KQ_d} \right], \\ \rho_L^{(l)} &= \Re \left[\sum_{u=1}^{Q_d} \sum_{k=1}^K e^{-j\pi \frac{K-1}{2K}} e^{j\pi \frac{\Psi_{(l-1)Q_d+u}^{(-1)}(k)-1}{K}} \frac{[\tilde{\mathbf{Z}}_d^{(l)}]_{k,u}}{KQ_d} \right]. \end{aligned} \quad (39)$$

The phase discrimination formula of the DLL is

$$ek_{\tau}^{(l)} = \frac{1}{2} \frac{\rho_E^{(l)} - \rho_L^{(l)}}{\rho_E^{(l)} + \rho_L^{(l)}}. \quad (40)$$

As shown in Figure 8, $d_{\tau} = \frac{1}{K\Delta f}$ is the equivalent spacing between the early and late branches, which corresponds to the correlation early-late spacing in the conventional code loop. d_{τ} is inversely proportional to the hopping bandwidth, i.e., the time resolution of DLL depends on the whole hopping bandwidth. Figure 9 shows the relation between DLL phase discrimination characteristics and the time delay after normalizing to $\frac{1}{K\Delta f}$. It can be seen that the discrimination characteristic curve for $K \geq 8$ is almost linear within $[\frac{-1}{K\Delta f}, \frac{1}{K\Delta f}]$. Note that there are discontinuities in Figure 9 at $[-3.5, -2.5, -1.5]/(K\Delta f)$ and $[1.5, 2.5, 3.5]/(K\Delta f)$. This is because in these cases $\rho_E^{(l)} = \rho_L^{(l)} = 0$, and denominator in (40) will equal to zero as a consequence.

As shown in Figure 6, $ek_{\tau}^{(l)}$ is the input for constant coefficient loop filter with transfer function as

$$F_{\tau}(s) = \frac{w_{\tau}}{K_{\tau}}, \quad (41)$$

where w_{τ} is the characteristic frequency and K_{τ} is the overall gain of the DLL. Since the output of DLL phase discriminator is the result of normalizing to $\frac{1}{K\Delta f}$, we have $K_{\tau} = K\Delta f$.

Similar to NCO_{pll} , the work of the NCO_{τ} is to transfer the time delay estimation result to compensation matrices. However, since NCO_{τ} only needs to calculate the compensation phase for hops intra-symbol, the structure of it is much simpler compared with NCO_{pll} . Adopting the common first-order integral structure to implement NCO_{τ} , the recursive form can be written as

$$\begin{aligned} \theta_{\tau}^{(l+1)} &= \theta_{\tau}^{(l)} + \frac{w_{\tau} Q_d T_s}{K_{\tau}} ek_{\tau}^{(l)}, \\ [\mathbf{P}_{\tau}^{(l)}]_{k,u} &= f_k^{((l-1)Q_d+u)} \theta_{\tau}^{(l)}. \end{aligned} \quad (42)$$

4.3 Closed-Loop Estimation Error Analysis

In the previous subsections, we have developed the structures and principles of PLL/DLL. Now we investigate the estimators corresponding to μ and τ in the

loop, and derive the RMSE of φ , μ , and τ estimation.

According to [34], aforementioned PLL can be characterized by a linearized equivalent model with additive noise as Figure 10. $\varphi_{pll}^{(l)}$, $\hat{\varphi}_{pll}^{(l)}$, and $\Delta\hat{\varphi}_{pll}^{(l)}$ denote the phase of input signal, the phase feedback adjustment, and the tracking error, respectively. $n_{pll}^{(l)}$ is a zero-mean Gaussian noise and its power spectral density normalized to the signal power is $\frac{N_0}{2A_{pll}^2}$ [34], where $\frac{N_0}{2}$ and A_{pll} are respectively the double-side noise power spectral density and the amplitude of PLL inputs. Considering the integral property of NCO_{pll} in (35) and (36), the transfer function corresponding to NCO_{pll} is equivalent to $F_{\text{NCO}_{pll}}(s) = \frac{1}{s}$. The system function of the PLL equivalent model can be expressed as

$$H_{pll}(s) = \frac{F_{pll}(s)}{s + F_{pll}(s)} = \frac{2\zeta w_{pll}s + w_{pll}^2}{w_{pll}^2 + 2\zeta w_{pll}s + s^2}. \quad (43)$$

Substituting $s = j2\pi f$ into (43), the equivalent noise bandwidth B_{pll} (half-side bandwidth) can be expressed as

$$B_{pll} = \int_0^{1/2} |H_{pll}(f)| df = Q_d T_s \frac{4\zeta^2 + 1}{8\zeta} w_{pll}, \quad (44)$$

where $Q_d T_s$ is the normalization coefficient. In the steady state, the error of $\hat{\varphi}_{pll}^{(l)}$ can be regarded as the transient integration of $n_{pll}^{(l)}$, and the variance of the phase error $\Delta\hat{\varphi}_{pll}^{(l)}$ is equal to the variance of $\hat{\varphi}_{pll}^{(l)}$. The variance of $\Delta\hat{\varphi}_{pll}^{(l)}$ is given by

$$\begin{aligned} \text{var}(\Delta\hat{\varphi}_{pll}^{(l)}) &= \text{var}(\hat{\varphi}_{pll}^{(l)}) \\ &\approx \text{var}(n_{pll}^{(l)}) \sum_{n=1}^l (2B_{pll})^n \\ &\xrightarrow{l \gg 1} \frac{2\text{var}(n_{pll}^{(l)}) B_{pll}}{(1 - 2B_{pll})}. \end{aligned} \quad (45)$$

Note that the equivalent model input is the combining result within $Q_d T_s$, we have

$$\frac{N_0}{2A_{pll}^2} = \frac{N_0}{2Q_d E_s} = \text{var}(n_{pll}^{(l)}), \quad (46)$$

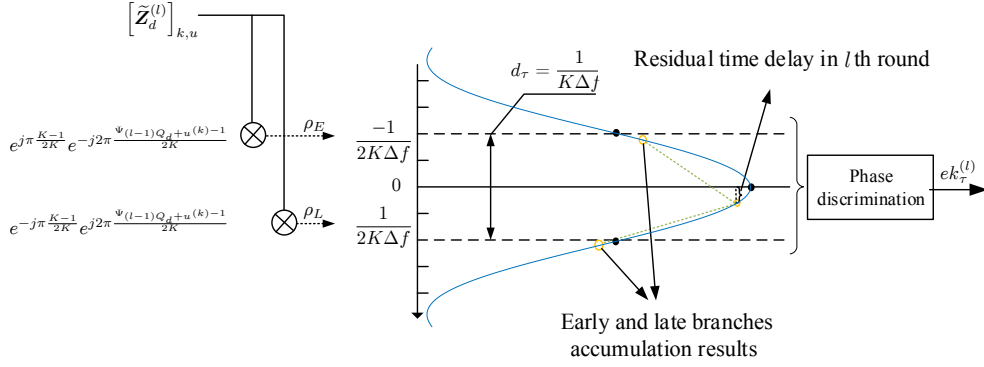


Figure 8. Block diagram of DLL phase discriminator.

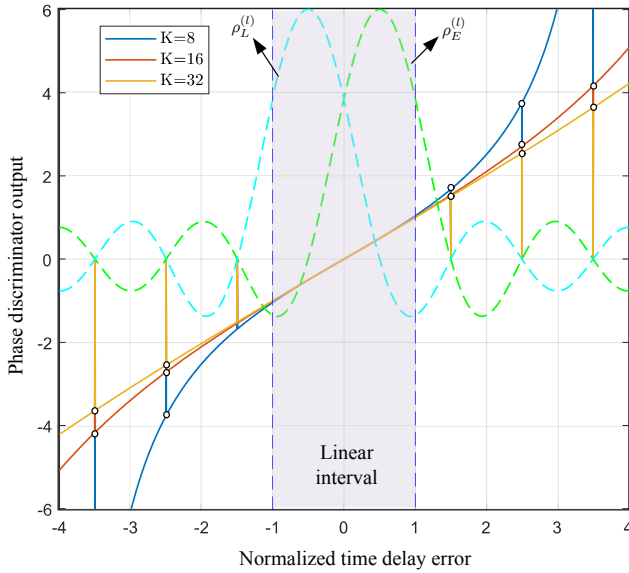


Figure 9. Plot of DLL phase discrimination characteristic.

where E_s denotes the symbol-level energy. By substituting (46) into (45), the RMSE of $\Delta\hat{\varphi}_{pll}$ can be written as

$$\sigma_{\Delta\hat{\varphi}_{pll}} \approx \sqrt{\frac{B_{pll}}{(1 - 2B_{pll})Q_d \frac{E_s}{N_0}}}. \quad (47)$$

As can be seen from (35) and (36), the output of the loop filter $\theta_{pll}^{(l)}$ corresponds to a frequency adjustment coefficient of input signal. From this perspective, $\theta_{pll}^{(l)}$ can be regarded as an estimate for μ . In the steady

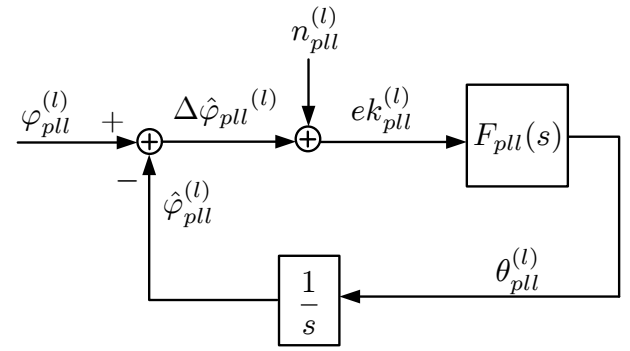


Figure 10. Equivalent model of PLL.

state, the estimate of μ is expressed as

$$\hat{\mu}_d^{(l)} = \frac{\theta_{pll}^{(l)}}{f_c}. \quad (48)$$

According to the form of $F_{pll}(s)$, by ignoring the correlation of the noise in the loop, the variance of $\theta_{pll}^{(l)}$ in the steady state is approximately expressed as

$$\text{var}(\theta_{pll}^{(l)}) \approx \left(\text{var}(n_{pll}^{(l)}) + \text{var}(\Delta\hat{\varphi}_{pll}^{(l)}) \right) \cdot (2\zeta w_{pll} + w_{pll}^2)^2. \quad (49)$$

By (48) and (49), the RMSE of $\hat{\mu}_d$ after normalizing to $Q_d T_s$ can be approximated as

$$\sigma_{\hat{\mu}_d} \approx \frac{1}{2\pi Q_d T_s f_c} (2\zeta w_{pll} + w_{pll}^2)$$

$$\sqrt{\frac{1}{(1 - 2B_{pl})2Q_d \frac{E_s}{N_0}}}. \quad (50)$$

For the estimation of time delay, $\theta_\tau^{(l+1)}$ is the filtered result of the DLL and can be directly used as an estimate of τ . To derive the RMSE, we first analyze the single-symbol case as with in the Section 4.2. In the presence of error noise, according to (38), the phase discrimination function of the u th symbol is expressed as

$$ek'_0 = \frac{1 \Re \left[ek_E e^{j\pi \frac{K-1}{2K}} \right] - \Re \left[ek_{LE} e^{-j\pi \frac{K-1}{2K}} \right] + \Re[n_E - n_L]}{2 \Re \left[ek_E e^{j\pi \frac{K-1}{2K}} \right] + \Re \left[ek_{LE} e^{-j\pi \frac{K-1}{2K}} \right] + \Re[n_E + n_L]}, \quad (51)$$

where n_E and n_L are the noises introduced by the early and late branches, respectively. Denote $\mathbf{n}_\tau$ as the noise vector corresponding to the u th symbol, n_E and n_L can be expressed as

$$\begin{aligned} n_E &= \frac{1}{K} \mathbf{n}_\tau^T \boldsymbol{\nu}_\tau, \\ n_L &= \frac{1}{K} \mathbf{n}_\tau^T \text{conj}(\boldsymbol{\nu}_\tau), \end{aligned} \quad (52)$$

where $\boldsymbol{\nu}_\tau = e^{-j2\pi \frac{(\mathbf{f}^{(u)} - \mathbf{f}_1)}{2K\Delta f}} \in \mathbb{C}^{K \times 1}$ and $\text{conj}(\cdot)$ represents the conjugate function. Note that $\boldsymbol{\nu}_\tau^H \boldsymbol{\nu}_\tau \rightarrow 0$ when K is sufficiently large. Therefore, n_E and n_L obey the independent zero-mean Gaussian distribution with variance $\text{var}(n_E) = \text{var}(n_L) = \frac{\sigma_n^2}{KA_\tau^2}$. A_τ denotes the hop-level signal amplitude of the u th symbol reception vector. Furthermore, let $w_1 = \Re[n_E - n_L]$ and $w_2 = \Re[n_E + n_L]$, then w_1 and w_2 obey the i.i.d. Gaussian distributions with variance $\text{var}(w_E) = \text{var}(w_L) = \frac{\sigma_n^2}{KA_\tau^2}$. In the steady state, the actual residual error λ in (37) is very small. Substituting (37) into (51) and taking an approximation, we can get

$$ek'_0 \xrightarrow{\lambda \rightarrow 0, K \gg 1} \frac{1}{2} \frac{w_1}{2\text{sinc}(\frac{1}{2}) + w_2}. \quad (53)$$

(53) shows that ek'_0 is approximately a $Z = \frac{X}{Y}$ random variable whose numerator and denominator are both Gaussian. According to [35, 36], a Gaussian distribution G_Z can be used to approximate above ran-

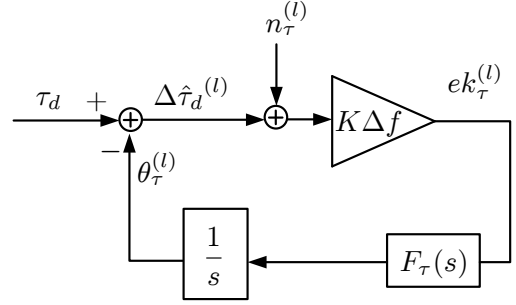


Figure 11. Equivalent model of DLL.

dom variable distribution in high SNR region. Denote the mean and root mean square of the numerator X and denominator Y as μ_X , σ_X , and μ_Y , σ_Y , respectively, then the mean value μ_Z and root mean square σ_Z corresponding to G_Z are

$$\begin{aligned} \mu_Z &= \frac{\mu_X}{\mu_Y}, \\ \sigma_Z &= \frac{\mu_X}{\mu_Y} \sqrt{\delta_X^2 + \delta_Y^2}, \delta_X = \frac{\sigma_X}{\mu_X}, \delta_Y = \frac{\sigma_Y}{\mu_Y}. \end{aligned} \quad (54)$$

Substitute $X = w_1$, $Y = 2\text{sinc}(\frac{1}{2}) + w_2$ and (54) into (53) yields

$$\Re[ek'_0] \sim \mathcal{N}\left(0, \frac{1}{16 \cdot \text{sinc}^2(\frac{1}{2})} \frac{\sigma_n^2}{KA_\tau^2}\right). \quad (55)$$

With above analysis of the PLL, an equivalent linearized model with additive noise for the DLL can be made as shown in Figure 11. The equivalent noise $n_\tau^{(l)}$ corresponds to $\Re[ek'_0]$ in (55). The system function of the equivalent model in Figure 11 is

$$H_\tau(s) = \frac{K_\tau F_\tau}{1 + K_\tau F_\tau} = \frac{w_\tau}{w_\tau + s}. \quad (56)$$

The equivalent noise bandwidth B_τ can be expressed as

$$B_\tau = \int_0^{1/2} |H_\tau(f)| df = Q_d T_s \frac{w_\tau}{4}. \quad (57)$$

Similar to the derivation process of (47), the single

symbol RMSE of $\hat{\tau}_d$ in the steady state can be obtained as

$$\sigma'_{\hat{\tau}_d} = \frac{1}{K\Delta f} \sqrt{\frac{B_\tau}{(1-2B_\tau)} \frac{1}{8 \cdot \text{sinc}^2(\frac{1}{2}) \frac{KA_s^2}{\sigma_n^2}}}, \quad (58)$$

where $\frac{1}{K\Delta f}$ is the factor from the loop error to the estimate of τ . For input signal matrices of the l th iteration, the analysis process is essentially the same as the single symbol, and the only difference is that the accumulation time is $Q_d T_s$. After using $\frac{E_s}{N_0}$ to represent SNR, the RMSE of $\hat{\tau}_d$ is expressed as

$$\sigma_{\hat{\tau}_d} = \frac{1}{K\Delta f} \sqrt{\frac{B_\tau}{(1-2B_\tau)} \frac{1}{8 \cdot \text{sinc}^2(\frac{1}{2}) Q_d \frac{E_s}{N_0}}}. \quad (59)$$

4.4 Decoder and Decoding Feedback

In the previous discussions about PLL/DLL, it is assumed $\mathbf{b}^{(l)} = \hat{\mathbf{b}}^{(l)} = \mathbf{1}_{Q_d}$. However, since information is mapped in the data segment, the symbols in $\mathbf{b}^{(l)}$ must be interleaved with $+1$ and -1 . In order to obtain the expectation of input symbols, the posterior LLR information $L_{pos}^{(l)}$ of the coded sequence should be calculated. Note that the code-type should be carefully selected so that the decoding LLR corresponds to the encoded sequence, such as Turbo, LDPC.

Considering the approximate ideal parameter compensation, the a posterior probability (APP) of the coded symbols $\mathbf{b}_c^{(l)}$ in the l th iteration is [37]

$$p(\mathbf{b}_c^{(l)} | \mathbf{Z}_d^{(l)}) \approx p(\mathbf{b}_c^{(l)} | \mathbf{Z}_d^{(l)}, \boldsymbol{\theta}^{(l)}), \quad (60)$$

where $\boldsymbol{\theta}^{(l)} = [\theta_{pll}^{(l)}, \theta_\tau^{(l)}]^T$, $\theta_{pll}^{(l)}$, $\theta_\tau^{(l)}$ correspond to the loop parameters in (34) and (42), respectively.

Assuming that each codeword has equal probability at the transmitter, then $\mathbf{b}_c^{(l)}$ satisfies

$$\mathbf{b}_c^{(l)} \in \mathcal{B}, p(\mathbf{b}_c^{(l)}) = \frac{1}{\text{card}(\mathcal{B})}, \quad (61)$$

where $\mathcal{B} \subset \{1, -1\}^{Q_c}$ is the code space, and function $\text{card}(A)$ yields the cardinality of set A . By performing hop-level combining on the coded symbols in $\dot{\mathbf{Z}}_d^{(l)}$, the soft information input into the decoder can be expressed as

pressed as

$$\boldsymbol{\xi}_{pri}^{(l)} = \Re \left[\mathbf{1}_K^T \cdot \left[\dot{\mathbf{Z}}_d^{(l)} \right]_{:,1:Q_c} \right]^T \in \mathbb{C}^{Q_c \times 1}. \quad (62)$$

For the u th coded symbol in the l th iteration, the posterior LLR information can be expressed as

$$L_{pos}^{(l)}(u) = \ln \left\{ \frac{p \left(\left[\mathbf{b}_c^{(l)} \right]_u = 1 | \boldsymbol{\xi}_{pri}^{(l)} \right)}{p \left(\left[\mathbf{b}_c^{(l)} \right]_u = -1 | \boldsymbol{\xi}_{pri}^{(l)} \right)} \right\}. \quad (63)$$

Substituting

$$p \left(\left[\mathbf{b}_c^{(l)} \right]_u = 1 | \boldsymbol{\xi}_{pri}^{(l)} \right) + p \left(\left[\mathbf{b}_c^{(l)} \right]_u = -1 | \boldsymbol{\xi}_{pri}^{(l)} \right) = 1$$

into (63), the posterior expectation of the u th symbol in the l th iteration is

$$\begin{aligned} \left[\boldsymbol{\xi}_{pos}^{(l)} \right]_u &= \mathbb{E} \left\{ \left[\mathbf{b}_c^{(l)} \right]_u \right\} \\ &= p \left(\left[\mathbf{b}_c^{(l)} \right]_u = 1 | \boldsymbol{\xi}_{pri}^{(l)} \right) - p \left(\left[\mathbf{b}_c^{(l)} \right]_u = -1 | \boldsymbol{\xi}_{pri}^{(l)} \right) \\ &= \tanh \left(\frac{e^{L_{pos}^{(l)}(u)}}{2} \right), \end{aligned} \quad (64)$$

where $\tanh(x) = \frac{e^{-x} - e^x}{e^{-x} + e^x}$ is a hyperbolic tangent function. Further, using $\boldsymbol{\xi}_{pos}^{(l)}$ as an estimate of $\mathbf{b}_c^{(l)}$ for feedback, (29) can be rewritten as

$$\hat{\mathbf{b}}^{(l)} = \left[\left[\boldsymbol{\xi}_{pos}^{(l)} \right]^T, \mathbf{1}_{Q_p}^T \right]^T. \quad (65)$$

V. CRLB OF PARAMETER ESTIMATION

In order to evaluate the performance of the proposed algorithm, we derive the CRLB of μ , τ , and φ in this section. The derivation process is based on the signal model of the open-loop stage in Section III.

Substitute the terms in the (12) with

$$\begin{aligned} \phi_\mu(u) &= 2\pi \left(f_c + \mathbf{f}^{(u)} \right) \\ \odot \left[\frac{T_h}{2} + [0 : K-1]^T T_h + (u-1)T_s \right] \mu &\in \mathbb{R}^{K \times 1}, \\ \phi_\tau(u) &= 2\pi \mathbf{f}^{(u)} \tau \in \mathbb{R}^{K \times 1}, \end{aligned}$$

$$\begin{aligned}\psi_\mu &= \text{vec}([\phi_\mu(1), \phi_\mu(2), \dots, \phi_\mu(Q_{pl})]) \in \mathbb{R}^{KQ_{pl} \times 1}, \\ \psi_\tau &= \text{vec}([\phi_\tau(1), \phi_\tau(2), \dots, \phi_\tau(Q_{pl})]) \in \mathbb{R}^{KQ_{pl} \times 1}, \\ \mathbf{g} &= e^{-j(\psi_\mu + \psi_\tau)} \in \mathbb{C}^{KQ_{pl} \times 1}.\end{aligned}\quad (66)$$

According to (10), the mean of the reception vector $\mathbf{z}_{pl}$ is the signal term $\mathbf{r}_{pl} = A_{\tau,\mu,\varphi}\mathbf{g}$. Its covariance matrix can be expressed as

$$\mathbf{\Gamma} = \begin{bmatrix} \sigma_n^2 \mathbf{I} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n^2 \mathbf{I} \end{bmatrix} \in \mathbb{C}^{KQ_{pl} \times KQ_{pl}}. \quad (67)$$

Let the vector of parameters to be estimated be

$$\boldsymbol{\theta} = [\mu, \tau, \varphi]^T. \quad (68)$$

The Fisher information matrix (FIM) [38] can be expressed as

$$\begin{aligned}[\mathbf{I}(\boldsymbol{\theta})]_{i,j} &= 2\Re \left\{ \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_i} \right]^H \mathbf{\Gamma}^{-1} \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_j} \right] \right\} \\ &\quad + \text{tr} \left[\mathbf{\Gamma}^{-1} \frac{\partial \mathbf{\Gamma}}{\partial [\boldsymbol{\theta}]_i} \mathbf{\Gamma}^{-1} \frac{\partial \mathbf{\Gamma}}{\partial [\boldsymbol{\theta}]_j} \right].\end{aligned}\quad (69)$$

Since $\boldsymbol{\theta}$ is independent of $\mathbf{\Gamma}$, the second term on the right of the equation in (69) is 0, then

$$\begin{aligned}[\mathbf{I}(\boldsymbol{\theta})]_{i,j} &= 2\Re \left\{ \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_i} \right]^H \mathbf{\Gamma}^{-1} \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_j} \right] \right\} \\ &= \frac{2}{\sigma_n^2} \Re \left\{ \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_i} \right]^H \left[\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_j} \right] \right\}.\end{aligned}\quad (70)$$

Using the chain rule to expand the partial derivative term in (70), it is derived that

$$\begin{aligned}\frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_1} &= \frac{\partial \mathbf{r}_{pl}}{\partial \mu} = \frac{\partial A_{\tau,\mu,\varphi}}{\partial \mu} \mathbf{g} + A_{\tau,\mu,\varphi} \frac{\partial \mathbf{g}}{\partial \mu}, \\ \frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_2} &= \frac{\partial \mathbf{r}_{pl}}{\partial \tau} = A_{\tau,\mu,\varphi} \frac{\partial \mathbf{g}}{\partial \tau}, \\ \frac{\partial \mathbf{r}_{pl}}{\partial [\boldsymbol{\theta}]_3} &= \frac{\partial \mathbf{r}_{pl}}{\partial \varphi} = \frac{\partial A_{\tau,\mu,\varphi}}{\partial \varphi} \mathbf{g}.\end{aligned}\quad (71)$$

By substituting (66) and (71) into (70), we have

$$\begin{aligned}\frac{\partial A_{\tau,\mu,\varphi}}{\partial \mu} &= \begin{cases} e^{-j\varphi} \frac{1}{\mu} [\cos(f_c T_h \mu) \\ -\text{sinc}(f_c T_h \mu)], & \mu \neq 0 \\ 0, & \mu = 0 \end{cases}, \\ \frac{\partial A_{\tau,\mu,\varphi}}{\partial \varphi} &= -j e^{-j\varphi} \text{sinc}(f_c T_h \mu), \\ \frac{\partial \mathbf{g}}{\partial \mu} &= -j \frac{\partial \psi_\mu}{\partial \mu} \odot \mathbf{g}, \\ \frac{\partial \mathbf{g}}{\partial \tau} &= -j \frac{\partial \psi_\tau}{\partial \tau} \odot \mathbf{g},\end{aligned}\quad (72)$$

where $\frac{\partial \psi_\mu}{\partial \mu}$ and $\frac{\partial \psi_\tau}{\partial \tau}$ correspond to the interval $1 \leq k \leq K$ and $1 \leq u \leq Q_{pl}$ are

$$\begin{aligned}\left[\frac{\partial \psi_\mu}{\partial \mu} \right]_{(u-1)K+k} &= 2\pi(f_c + f_k^{(u)}) \\ &\quad \cdot \left[\frac{T_h}{2} + (k-1)T_h + (u-1)T_s \right], \\ \left[\frac{\partial \psi_\tau}{\partial \tau} \right]_{(u-1)K+k} &= 2\pi f_k^{(u)}.\end{aligned}\quad (73)$$

By combining (70), (72) and (73), the estimation CRLB of μ, τ, φ can be derived from the corresponding diagonal elements of the inverse matrix of FIM [38] as

$$\begin{aligned}\text{CRLB}(\mu) &= [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{1,1}, \\ \text{CRLB}(\tau) &= [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{2,2}, \\ \text{CRLB}(\varphi) &= [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{3,3}.\end{aligned}\quad (74)$$

VI. SIMULATIONS AND DISCUSSIONS

In this section, simulation results are provided to illustrate the performance of the proposed algorithms. Suppose the normalized symbol rate $R_s = 1$ sps. Without loss of generality, we set the center frequency of the received signal to $f_c = 10^7 R_s$. $K = 32$ and $M = 8$ are adopted in simulation, respectively, and thus, the frequency space $\Delta f = MKR_s = 256R_s$. In addition, it should be noting that the CRLB in the following figures correspond to the RMSE results of the estimation.

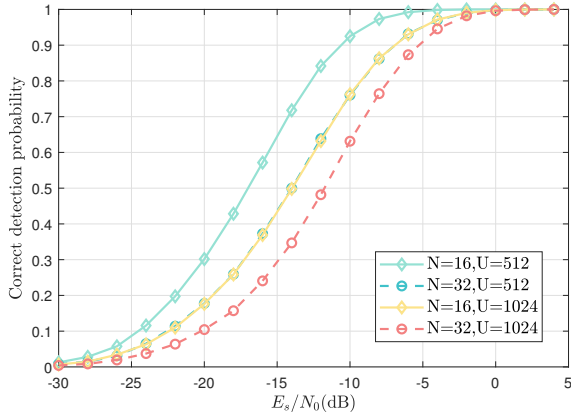


Figure 12. Correct detection probability of open-loop search.

For the open-loop stage, the length of the pilot sequence is set to $Q_{pl} = 2048$. To show the impact of different search intervals of μ and τ , the parameters are selected as $N \in \{16, 32\}$, $U \in \{512, 1024\}$. Particularly, when $N = 16$ and $U = 512$, the true values of μ and τ are assumed to traverse the search range in steps of $\frac{0.01}{16f_c Q_{pl} T_s}$ and $\frac{0.01}{16K \Delta f}$, which means that 10^4 test points for each search grid. For all the simulations, 1000 times of Monte Carlo simulation are performed for each test point.

Figure 12 demonstrates the correct detection probability of the grid-based open-loop search algorithm. It is clear that the correct detection probability of open-loop search varies with search parameters. For the same SNR, the detection probability is the highest in the case of $N = 16$, $U = 512$ and lowest in the case of $N = 32$, $U = 1024$. The two cases of $N = 16$, $U = 1024$ and $N = 32$, $U = 512$ have almost the same correct detection probability. The main reasons are as follows, firstly, the detection becomes more sensitive to noise as the search grid tends to be denser with the gap between the mean values of adjacent grid points getting smaller; secondly, the peak shape is sinc roll-off in both search dimensions of μ and τ for positions close to the true value. Therefore, keeping the search precision of one dimension unchanged but increasing that of the other dimension by the same proportion results in approximately the same effect on the overall detection performance.

Figure 13 shows the RMSE of μ , τ , and φ under correct detection. It can be seen that the RMSE of μ and φ are only sensitive to N , while the RMSE of τ

is only sensitive to U . This corresponds to the results in (25) and (26). It is worthy noting that the RMSE curves of μ , τ , and φ tend to be flat at high SNR, and the gap with CRLB increases. This is because of the discrete nature of grid-based search. When the true value of the parameter is not aligned with the points of the search grid, the peak will appear on the point closest to the true value with high SNR. As a result, the increase in SNR at this point will not reduce the error. Taking the μ -dimension search as an example, when the SNR is high, neglecting the effect of noise, the true values lying within the interval of $\frac{1}{2NQ_{pl}f_c T_s}$ apart from a specific grid point p_μ will be treated as p_μ . The simulation RMSE result can be approximated as

$$\text{RMSE}(\mu)_{E_s/N_0 \rightarrow \infty} \approx \sqrt{\frac{N}{800} \sum_{i=0}^{\frac{16 \times 100}{2N}} \left(\frac{0.01}{16Q_{pl}f_c T_s} i \right)^2}. \quad (75)$$

The upper limit of the summation in (75) is the number of test points within $\left[p_\mu - \frac{1}{2NQ_{pl}f_c T_s}, p_\mu + \frac{1}{2NQ_{pl}f_c T_s} \right]$.

Similarly, the simulation RMSE results for τ , φ can be approximated as

$$\begin{aligned} \text{RMSE}(\tau)_{E_s/N_0 \rightarrow \infty} &\approx \sqrt{\frac{U}{800K} \sum_{i=0}^{\frac{16 \times 100K}{2U}} \left(\frac{0.01}{16K \Delta f} i \right)^2}, \\ \text{RMSE}(\varphi)_{E_s/N_0 \rightarrow \infty} &\approx \sqrt{\frac{N}{800} \sum_{i=0}^{\frac{16 \times 100}{2N}} \left(\frac{0.01\pi}{16} i \right)^2}. \end{aligned} \quad (76)$$

The red dotted lines in Figure 13 demonstrates the results of substituting the simulation parameters into (75) and (76).

For the closed-loop stage, the LDPC code with code rate $\frac{1}{2}$ and code length $Q_c = 128$ is adopted. On this condition, the relationship between the bit-level SNR and the symbol-level SNR is $[E_b/N_0]_{\text{dB}} \approx [E_s/N_0]_{\text{dB}} + 3$ dB. The length of the padding sub-segment is $Q_p = 2$, i.e., the length of one segment is $Q_d = Q_c + Q_p = 130$. We assume that the errors of parameter input to the closed-loop algorithm correspond to the maximum residual of grid-based open-loop search stage. Substituting

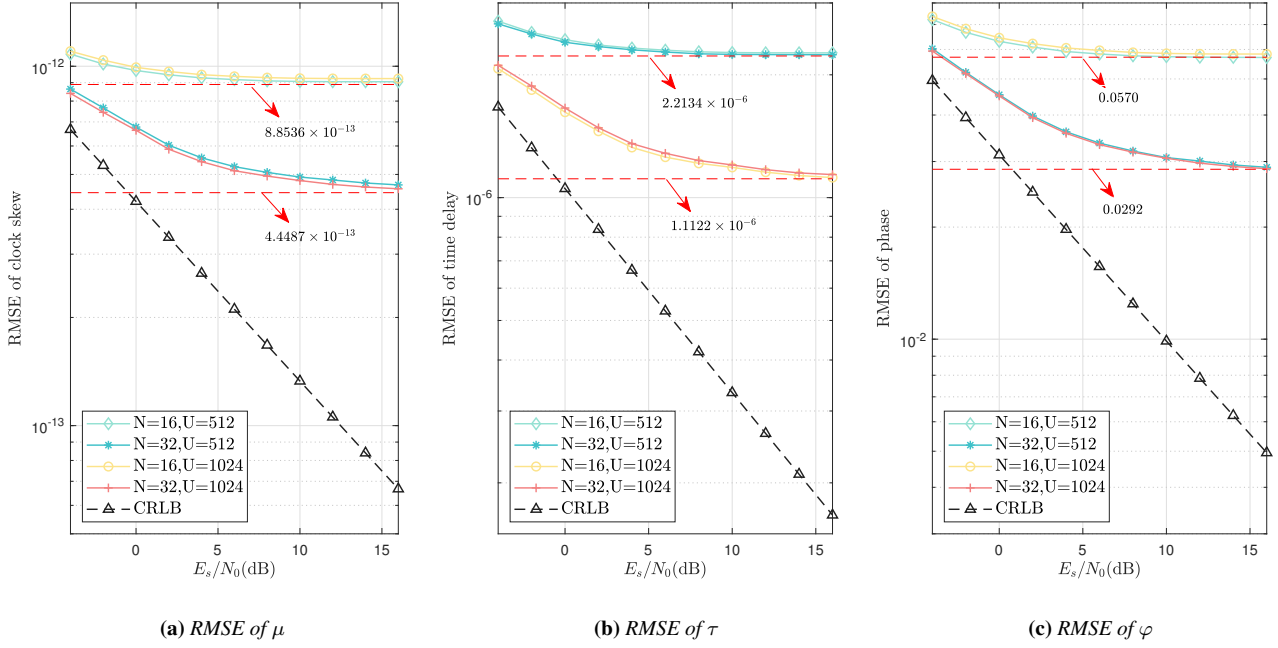


Figure 13. RMSE of grid-based open-loop search.

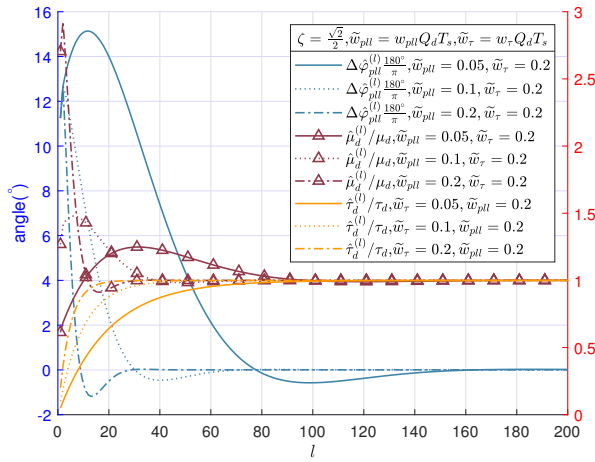


Figure 14. Plot of iteration convergence.

$N = 16, U = 16K$ into (25) and (26) yields the residual error $\mu_d = \frac{1}{NQ_{pl}f_cT_s}$, $\tau_d = \frac{1}{U\Delta f}$, $\varphi_d = \frac{\pi}{16}$.

Figure 14 shows the converge curves of $\Delta\hat{\varphi}_{pll}^{(l)}$, $\hat{\mu}_d^{(l)}$, $\hat{\tau}_d^{(l)}$. Denote the $\frac{1}{Q_dT_s}$ -normalized characteristic frequency as $\tilde{w}_{pll} = w_{pll}Q_dT_s$ and $\tilde{w}_\tau = w_\tau Q_dT_s$, and let $\zeta = \frac{\sqrt{2}}{2}$ (the optimal damping factor for a second-order phase-locked loop with a frequency step response is $\zeta_{opt} = \frac{\sqrt{2}}{2}$ [32, 39]). Note that the curves

of $\hat{\mu}_d^{(l)}$ and $\hat{\tau}_d^{(l)}$ are the results of normalization to the true value. It can be seen that PLL and DLL enter a stable state after about 160 iterations. The $\hat{\mu}_d^{(l)}$ and $\hat{\tau}_d^{(l)}$ eventually approach to the true value, and the $\Delta\hat{\varphi}_{pll}^{(l)}$ tend to zero. It can be seen that the performances converge rapidly, which makes our proposed algorithms suitable for practical applications.

Let $v_{\hat{\mu}_d}$, $v_{\hat{\tau}_d}$ and $v_{\Delta\hat{\varphi}_{pll}}$ denote the RMSE of the μ_d , τ_d , φ_d estimation in the steady state of simulation, respectively. Then, $v_{\hat{\mu}_d}$, $v_{\hat{\tau}_d}$, and $v_{\Delta\hat{\varphi}_{pll}}$ versus SNR are shown in Figure 15 (a), (b), and (c). The theoretical curves of RMSE for comparison in the Figure 15 correspond to (47), (50), (59). It can be seen that when $E_b/N_0 > 3$ dB, the simulation curves of the RMSE is approximately consistent with the theoretical results. When E_b/N_0 is lower than 0 dB, the results of $v_{\hat{\tau}_d}$, and $v_{\Delta\hat{\varphi}_{pll}}$ are significantly higher than the theoretical results. This is because there are a number of symbol polarity inversions caused by decoding errors in the $\xi_{pos}^{(l)}$ when the SNR is low, and the inputs of the PLL and DLL discriminator will no longer be close to the coherent combining result of Q_d symbols. Figure 15(d) shows the bit error rate (BER) of the loop in the stable state and the loop with ideal compensation of μ_d , τ_d , φ_d . and the simulation

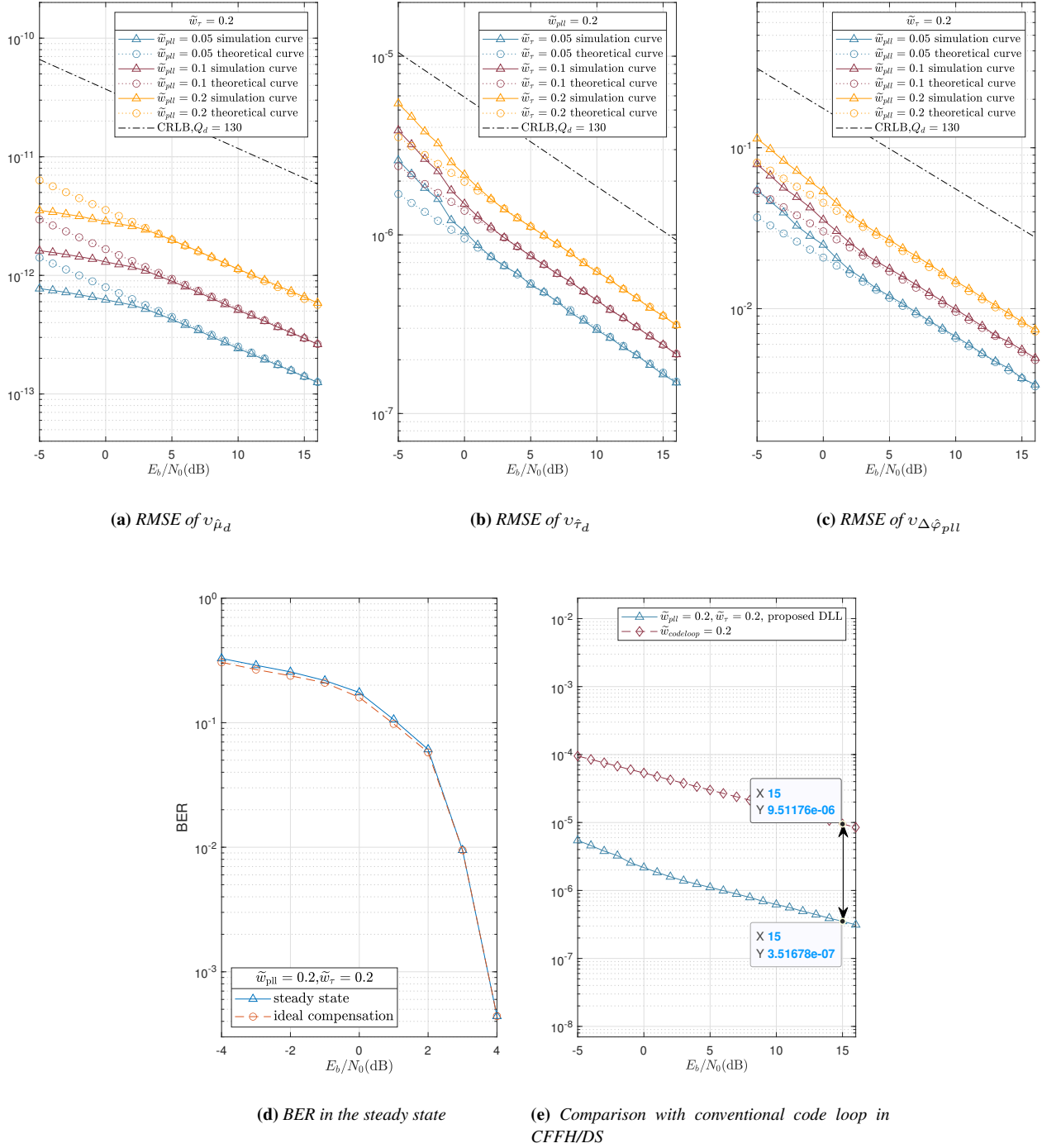


Figure 15. Closed-loop simulation results.

results of $\xi_{pos}^{(l)}$ decision. It can also be noted that the results of $v_{\hat{\mu}_d}$ are smaller than the theoretical results in the E_b/N_0 interval of $[-4, 0]$ dB. The reason is that the output of decoder approximates the real part of the input noise at a low SNR, and the PLL's phase dis-

crimination output will remain concentrated around 0, rather than increasing with the noise variance. In addition, by comparing the RMSE in Figure 13, it can be seen that for the high SNR condition, when the normalized characteristic frequency is set to $\tilde{w}_{pll} = 0.05$

and $\tilde{w}_\tau = 0.05$, the closed-loop estimation RMSE of time delay, clock skew, and carrier phase would be significantly smaller than that of the open-loop grid-based search estimation at the same SNR ratio in the steady state. In order to evaluate the μ , τ estimation, and phase tracking performance in the steady state, the CRLB results of Q_d symbols pilot sequence are also presented in Figure 15(a)(b)(c). It can be seen that for logarithmic y-coordinates, when E_b/N_0 is greater than 3 dB, the curves of $v_{\hat{\mu}_d}$, $v_{\hat{\tau}_d}$, and $v_{\Delta\hat{\varphi}_{pll}}$ are parallel to the curves of corresponding CRLB. Further, by numerical calculation, the interval between $v_{\hat{\mu}_d}$, $v_{\hat{\tau}_d}$, $v_{\Delta\hat{\varphi}_{pll}}$ results and the corresponding CRLB can be approximated as (77) with $E_b/N_0 > 5$ dB.

$$\begin{aligned} v_{\hat{\mu}_d}(E_b/N_0) &\approx 0.2890 \cdot \left(2\zeta w_{pll} + \frac{w_{pll}^2}{2} \right), \\ &\quad \cdot \sqrt{\frac{1}{(1-2B_{pll})}} \text{CRLB}_{\hat{\mu}}(E_b/N_0) \\ v_{\hat{\tau}_d}(E_b/N_0) &\approx \sqrt{\frac{B_\tau}{(1-2B_\tau)}} \text{CRLB}_{\hat{\tau}}(E_b/N_0), \\ v_{\Delta\hat{\varphi}_{pll}}(E_b/N_0) &\approx 0.5 \cdot \sqrt{\frac{1}{(1-2B_{pll})}} \text{CRLB}_{\hat{\varphi}}(E_b/N_0). \end{aligned} \quad (77)$$

Figure 15(e) compares the time delay estimation RMSE of the proposed algorithm and the code loop algorithm of CFFH/DS system in [14]. Considering the fair comparison, the CFFH/DS system is set to the same symbol rate, order of loop, and length of accumulation time in the present simulation. In addition, the CFFH / DS system adopts symbol-by-symbol coherent integration in the simulation. As can be seen, for the same normalized characteristic frequency $\tilde{w}_{code\ loop} = \tilde{w}_\tau = 0.2$, the RMSE results of proposed algorithm is significantly lower than that of the code loop algorithm within a wide SNR interval. By comparing the simulation results, the RMSE estimated by the proposed algorithm is only about $\frac{1}{27}$ of the code loop algorithm at $E_b/N_0 = 15$ dB.

VII. CONCLUSION

In this paper, we proposed a novel fine synchronization algorithm for CFFH systems. The algorithm first locks

the parameter errors within a small range by the grid-based open-loop search process, followed by further precise estimation of the parameters using the closed-loop estimation method. This paper presented a detailed implementation structure for each stage and analyzed the error of the estimates. In addition, this paper derived the RMSE of estimate of time delay, clock skew, and carrier phase in closed-loop stage, as well as the CRLB. The simulation results showed that the system can achieve good tracking of the clock skew and carrier phase, while the time delay estimation has a significant improvement compared to the conventional code loop algorithm.

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Biographies



Xuanhe Yang received the bachelor's degree from the School of Information Engineering, Zhengzhou University, China, in 2015. His current research is focused on spread spectrum signal processing, satellite communication, IoT technology, and physical-layer security.



Chang Li received the B.Eng. degree from School of Information and Electronics, Beijing Institute of Technology (BIT), China in 2022. Now, she is pursuing a M.Sc. degree in School of Cyberspace Science and Technology, BIT. Her research interests include satellite communication and physical-layer security.



Weike Zhang received the Ph.D. degree from National University of Defense Technology in 2019. His current research interests mainly include array signal processing, radar electronic countermeasures and data link.



Xiaqing Miao assistant professor, received the Ph.D. degree at the Beijing Institute of Technology, Beijing, China in 2020. His research interests include wireless communication and signal processing.



Shixun Luo received the B.Eng. and Ph.D. degrees from the Beijing Institute of Technology (BIT), Beijing, China, in 2013 and 2020 respectively. Since 2020, she has been with the School of Cyberspace Science and Technology BIT, where she is currently working as an experimentalist. Her research interests include satellite communication, UAV network, and physical-layer security, especially spread spectrum signal processing.



Aihua Wang (M'08) received the B.Eng. degree from Hebei University, Baoding, China, in 1986, the M.Sc. degree from Xidian University, Xi'an, China, in 1993, and a Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2001, respectively, all in communications and information systems. She now holds the post as a Full Professor with the School of Information Science and Electronics, Beijing Institute of Technology. She has published more than 40 journal and conference papers. She also holds (or co-holds) more than 20 patents. Her main research interest includes nonorthogonal multiple access, nonterrestrial network, satellite communications, and machine learning.